Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian Variational Ansatz

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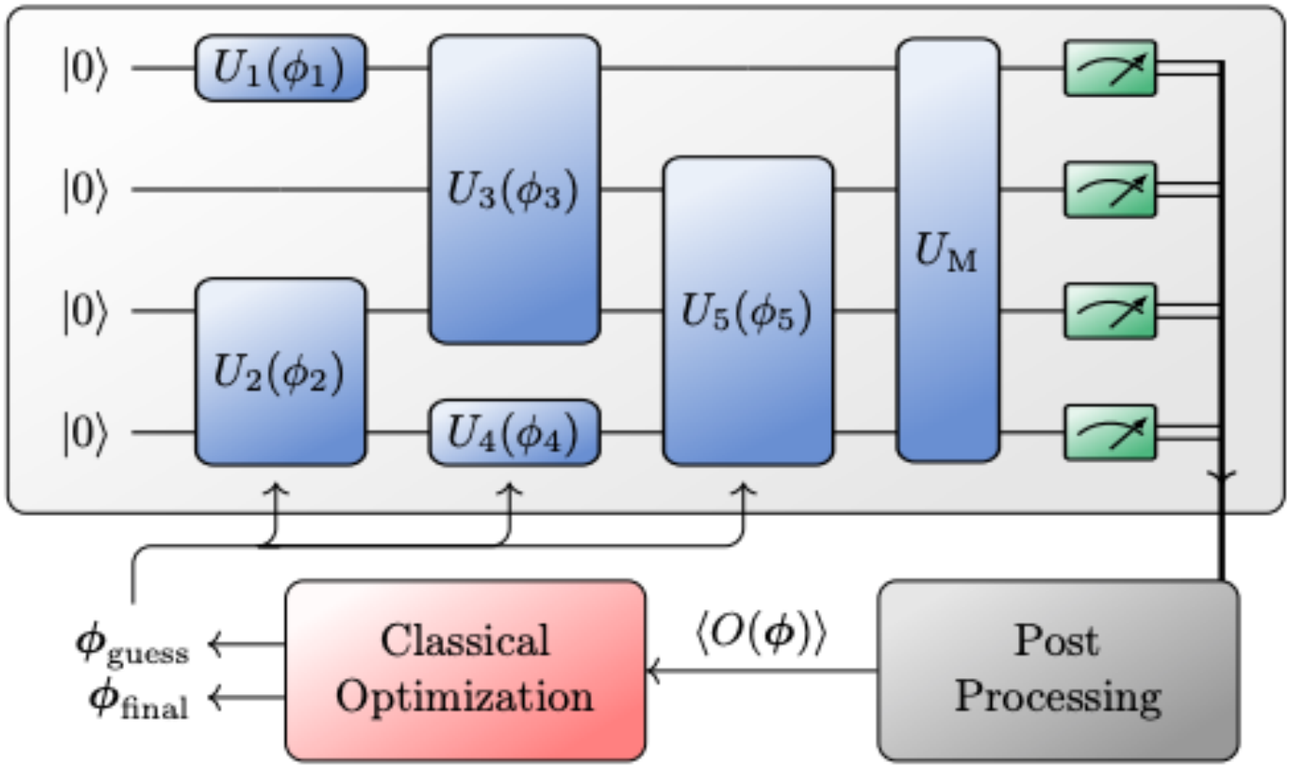
Variational Quantum Algorithms

- Leading **NISQ strategy**

- The problem is encoded in **minimising a cost function**

(e.g. finding Ground state of an Hamiltonian)

(e.g. Hamiltonian expectation value)



VQAs scheme [Bittel et al., PRL (2021)]

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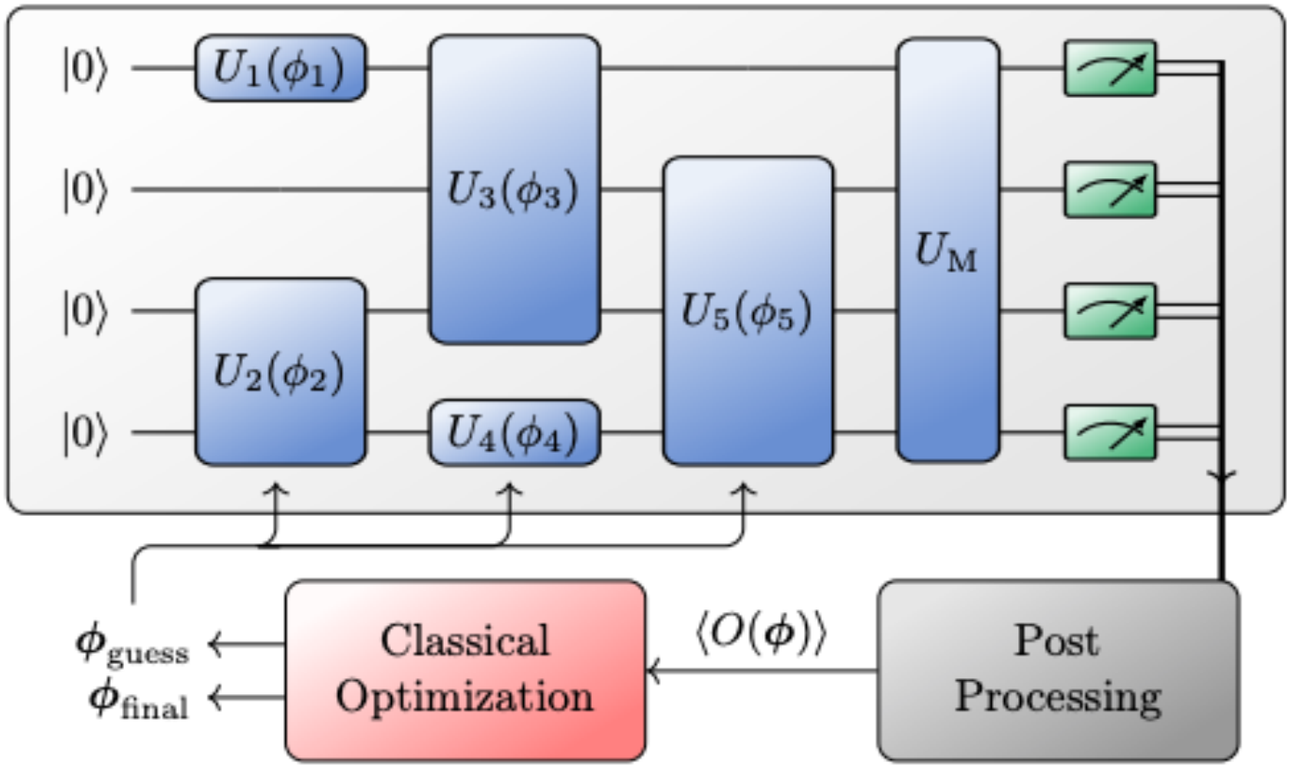
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3. Classical **optimization**



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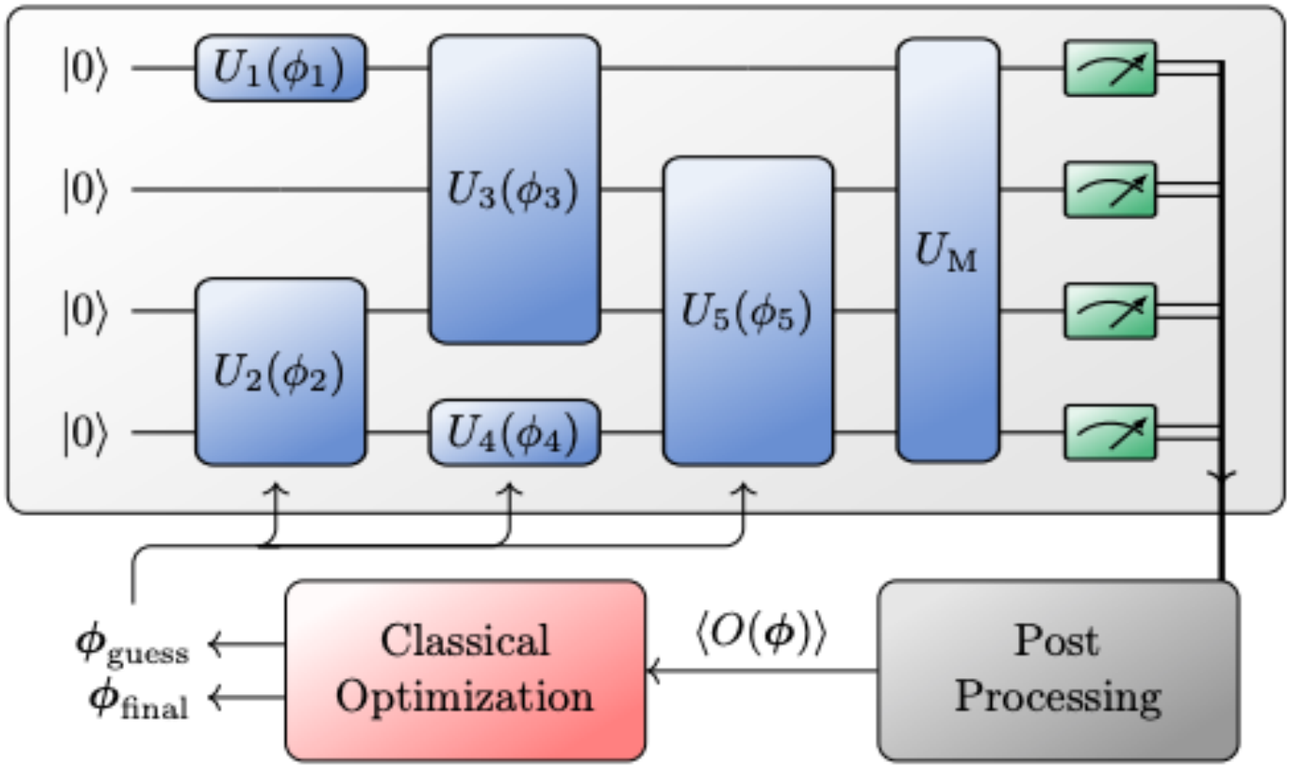
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MAIN DIFFICULTIES:

- **Noise**
- **Non-convex** optimization
- Flat landscape (a.k.a. **Barren Plateaus**)



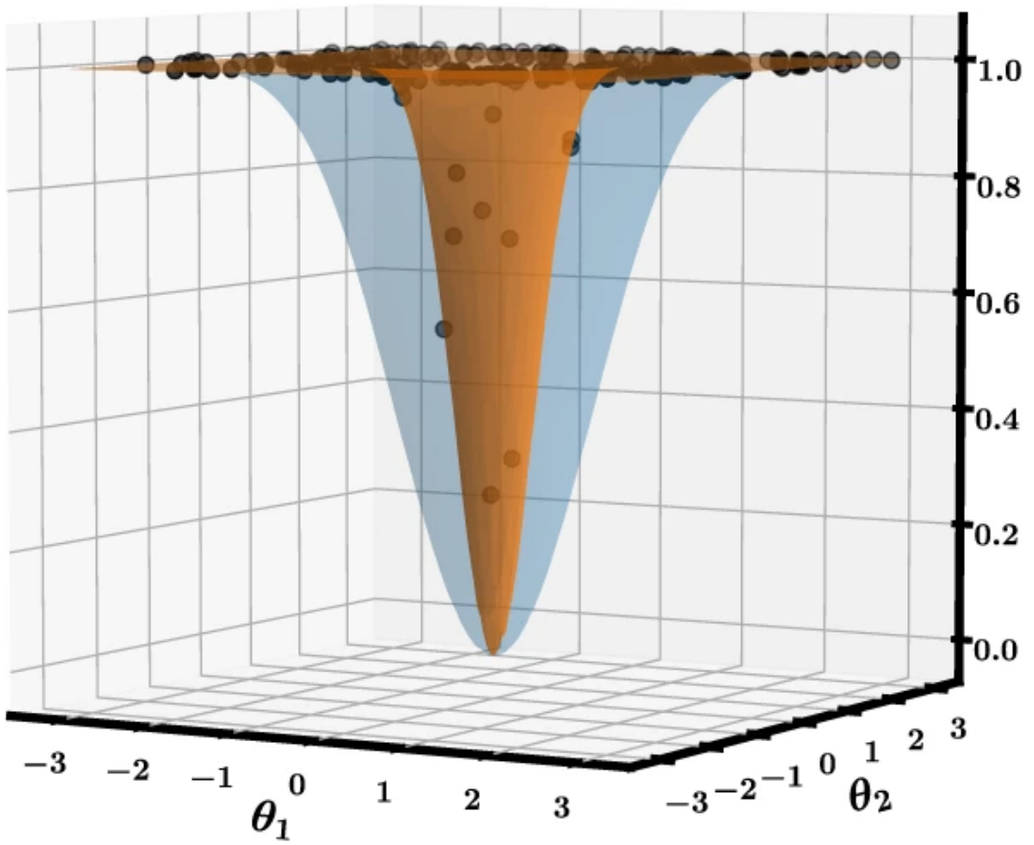
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Barren Plateaus

High circuit expressibility → **Exponential vanishing gradients with number of qubits N**

(Barren Plateaus definition)

[McClean et al., Nat. Comm. (2018)]
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Orange (blue) landscape N = 24
(N = 4) qubits. [Cerezo et al.,Nature(2021)]

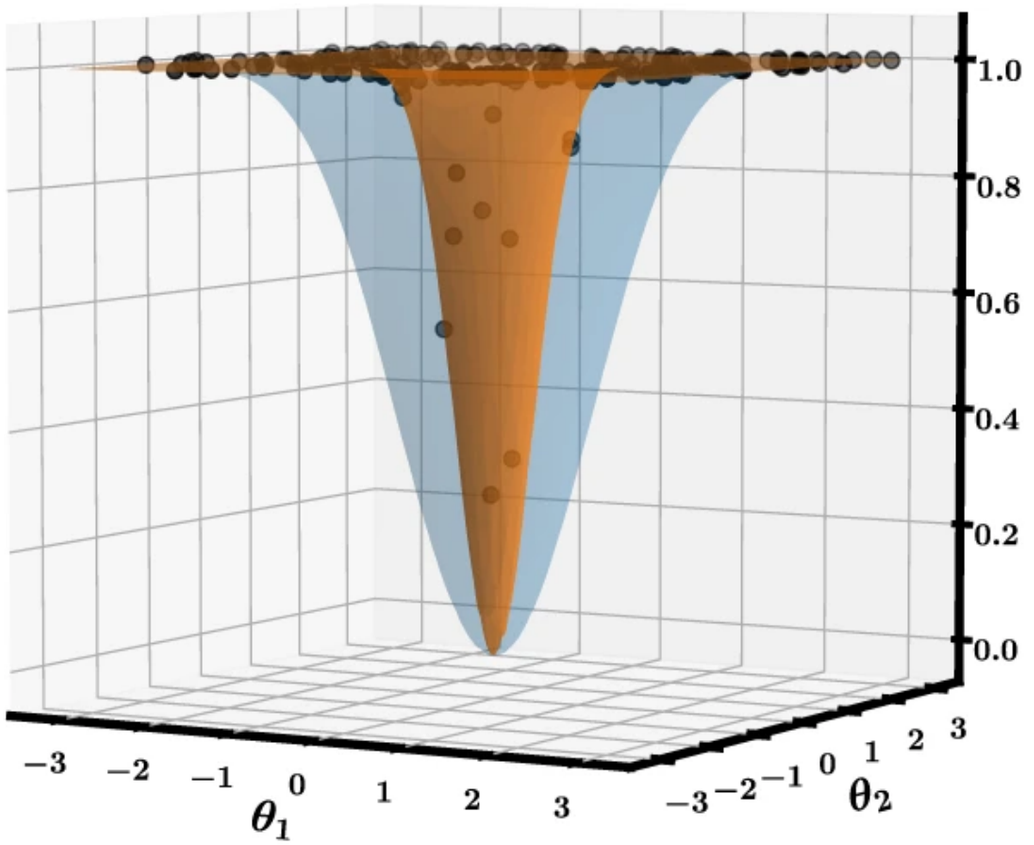
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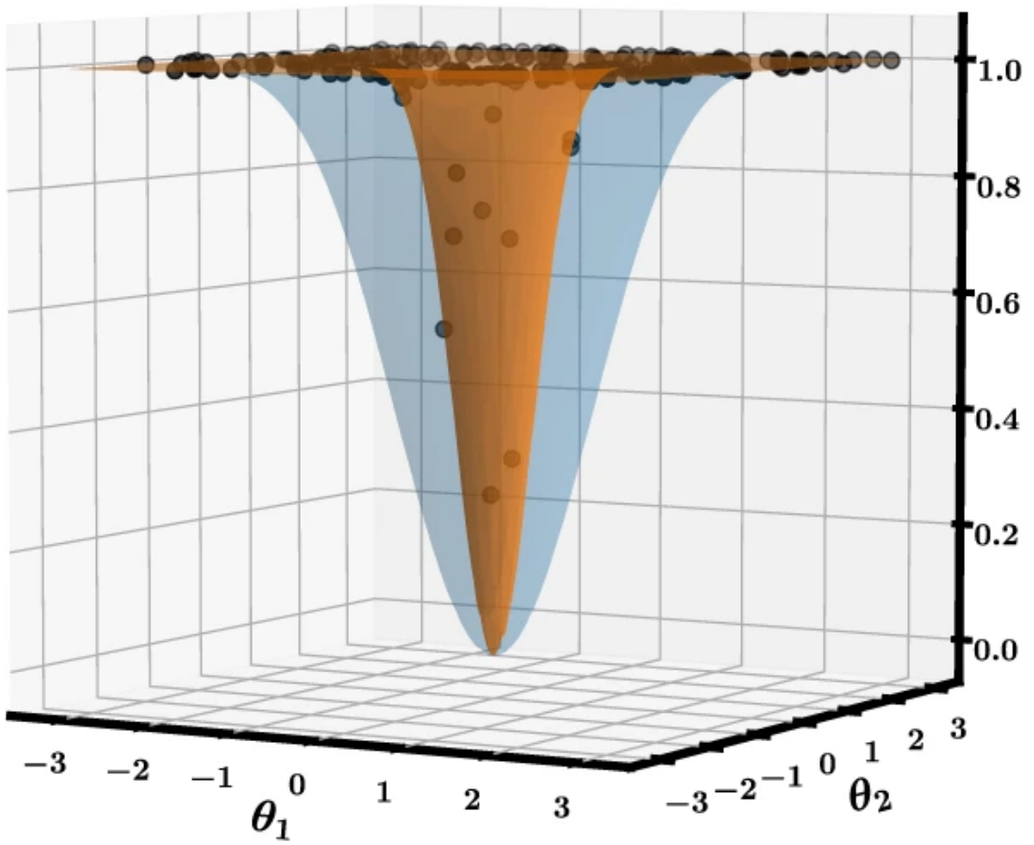
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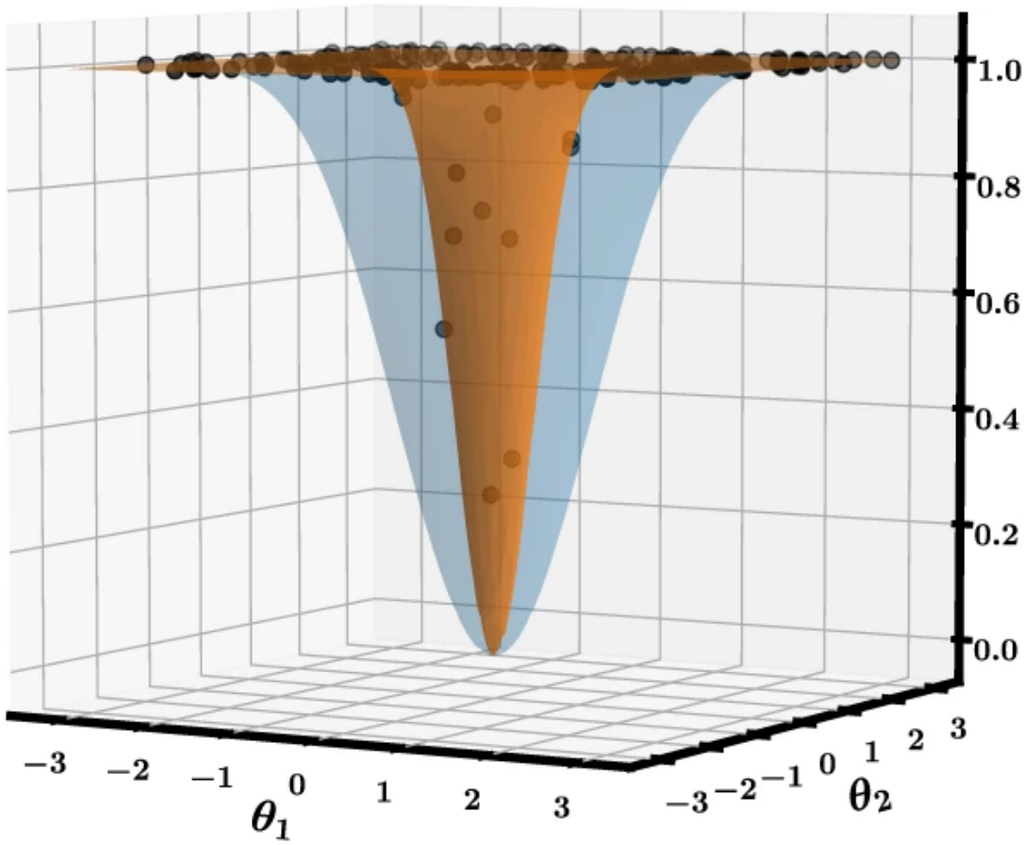
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☠ **Serious problem for VQAs** ☠



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Problem-inspired Ansatz

$$|\psi(\boldsymbol{\gamma})\rangle = \prod_{m=1}^{P} e^{-i\gamma_{m,M}H_M} \cdots e^{-i\gamma_{m,1}H_1} |\psi_0\rangle$$

The problem **Hamiltonian** is a **linear combination of these generators**

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$$H_{\mathrm{XYZ}} = \sum_{j=1}^{N} \left(X_j X_{j+1} + \Delta_Y Y_j Y_{j+1} + \Delta_Z Z_j Z_{j+1} \right)$$

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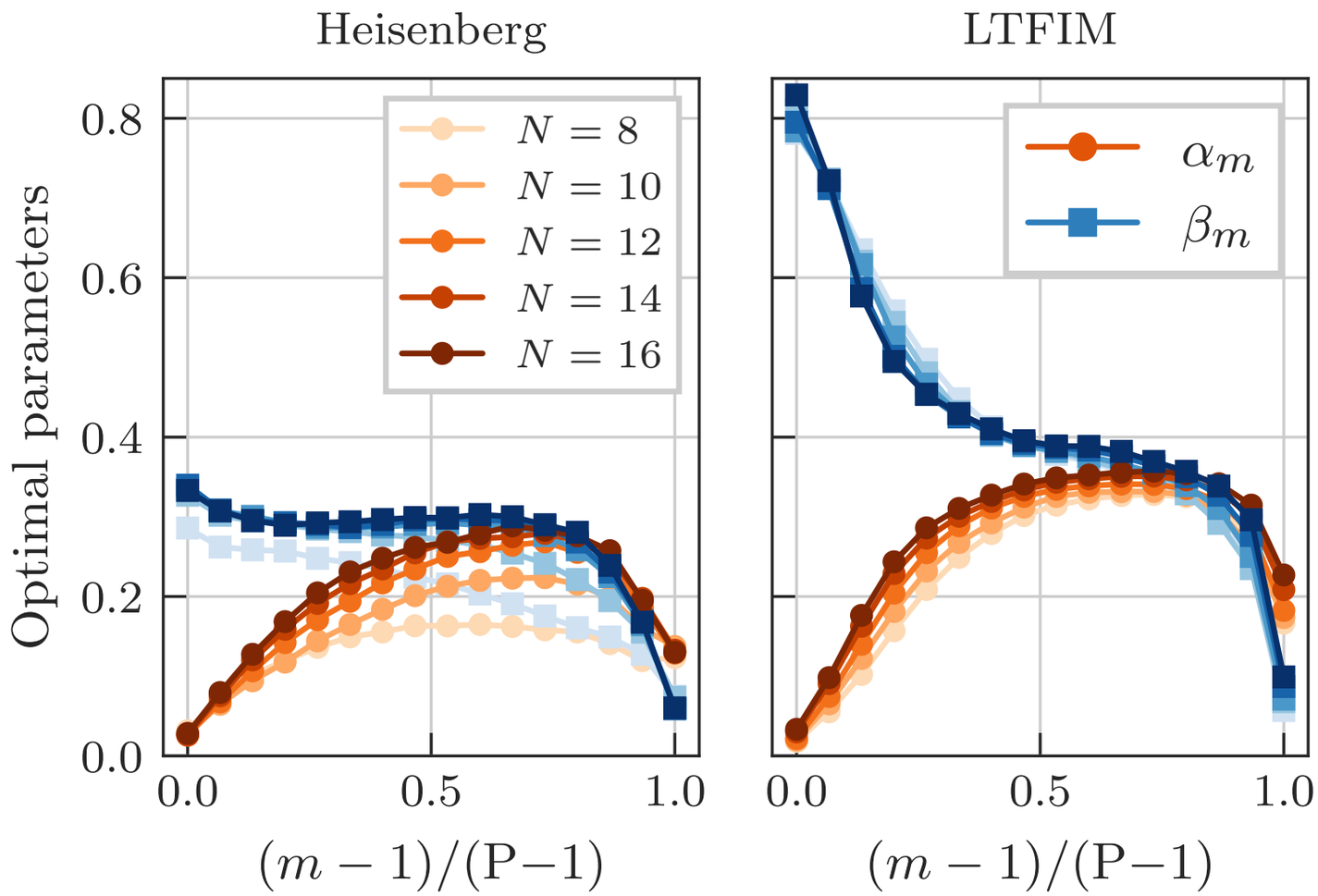
$$H_{\mathrm{LTFIM}} = \sum_{j=1}^{N} Z_jZ_{j+1} - g_x\sum_{j=1}^{N} X_j - g_z\sum_{j=1}^{N} Z_j$$

Although symmetry-ansatz, there can be Barren Plateaus

[Larocca et al., ArXiv (2021)]

Pattern in Optimal Parameters

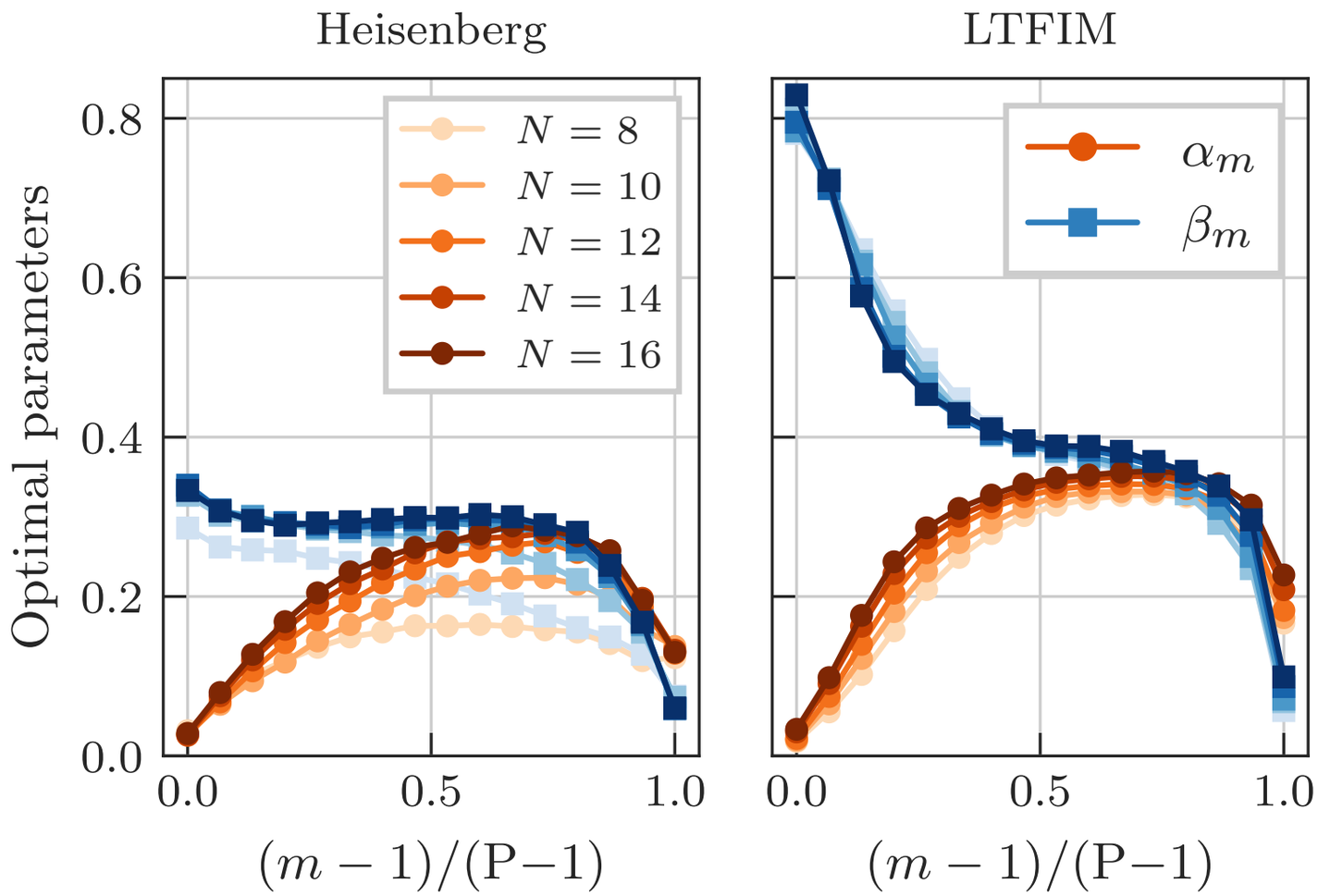
$$(\alpha_1, \ldots, \alpha_P, \beta_1, \ldots, \beta_P,)$$



(Rescaled) index with which we labeled the parameters

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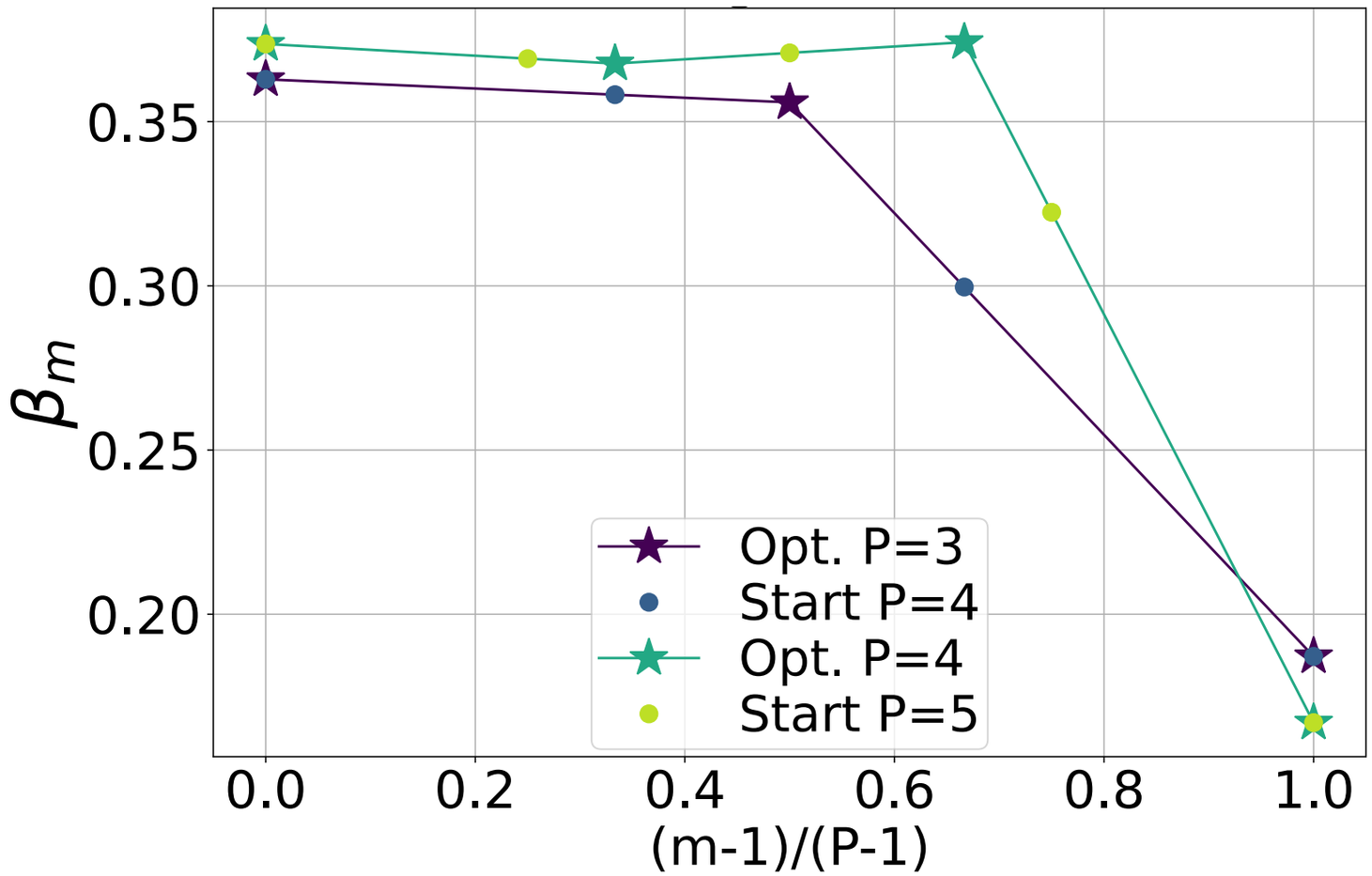


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How can we find this pattern?

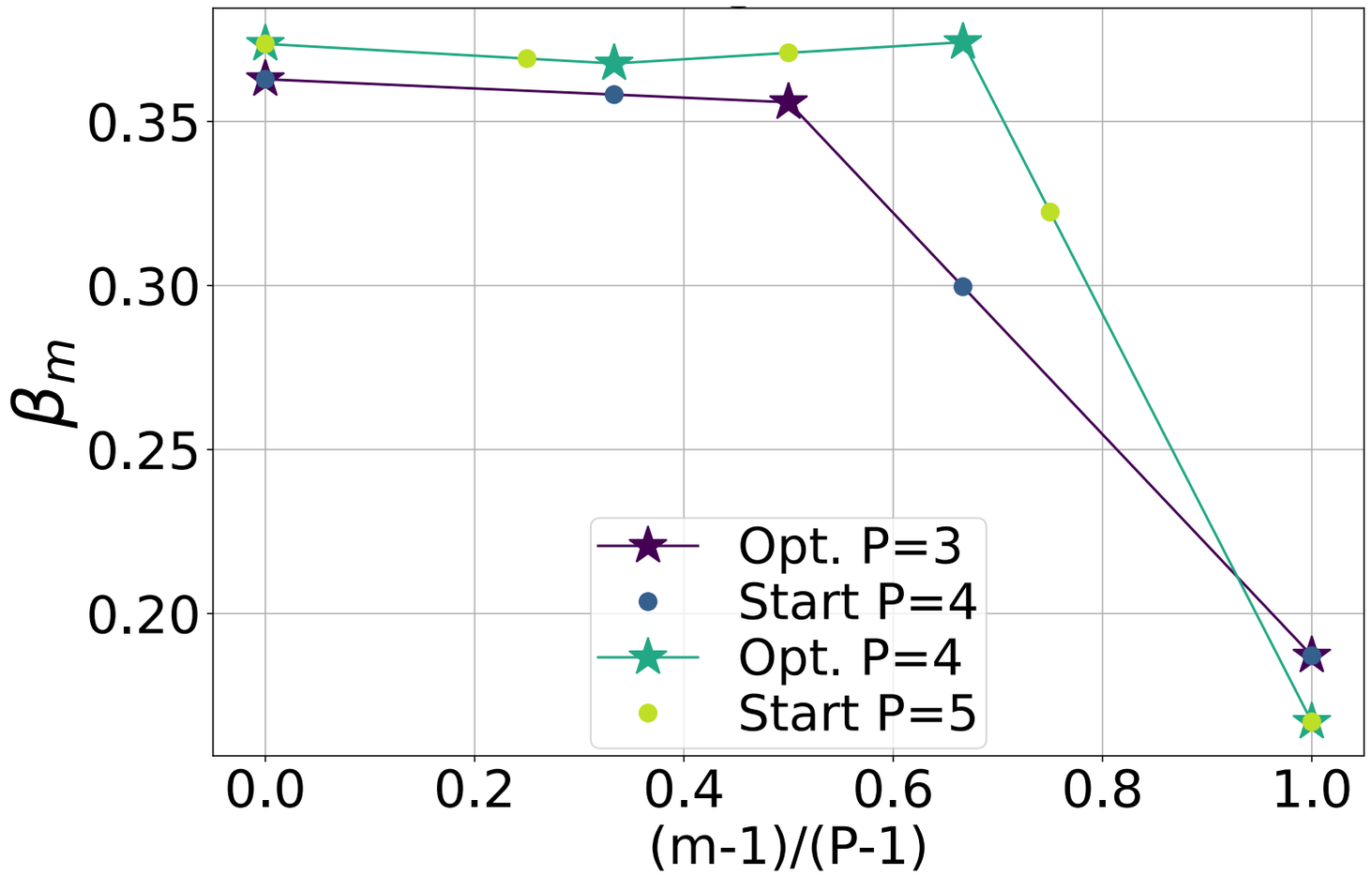
INTERP strategy

[Zhou et al., PRX. (2018)]

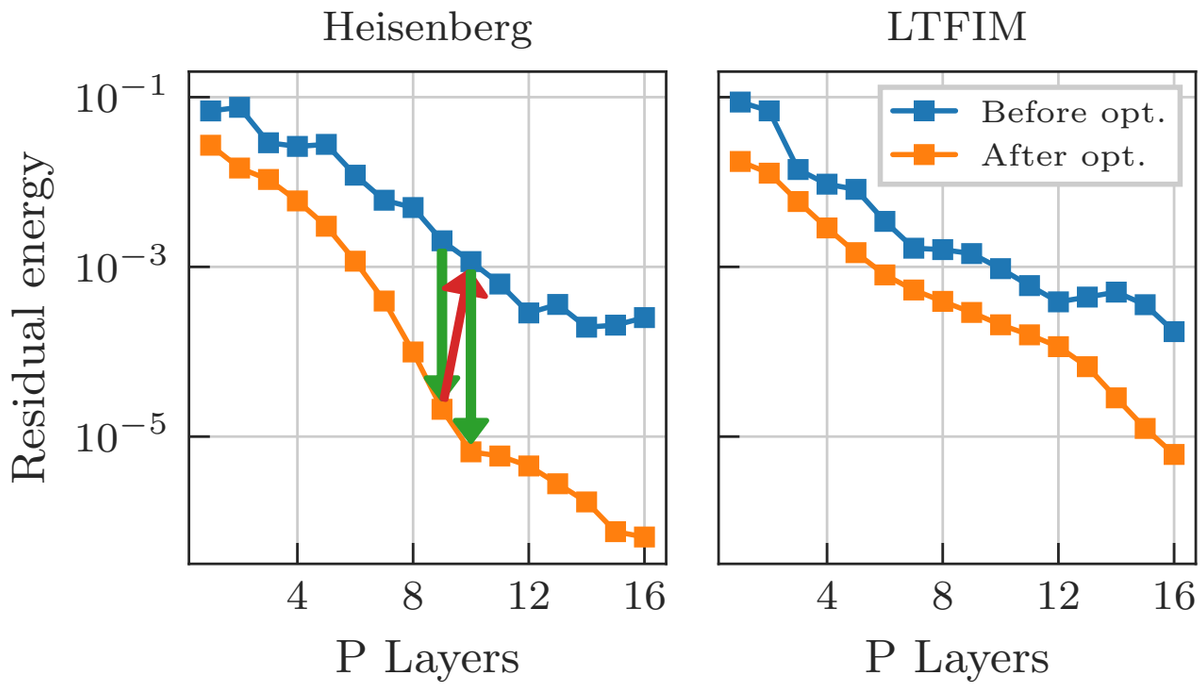


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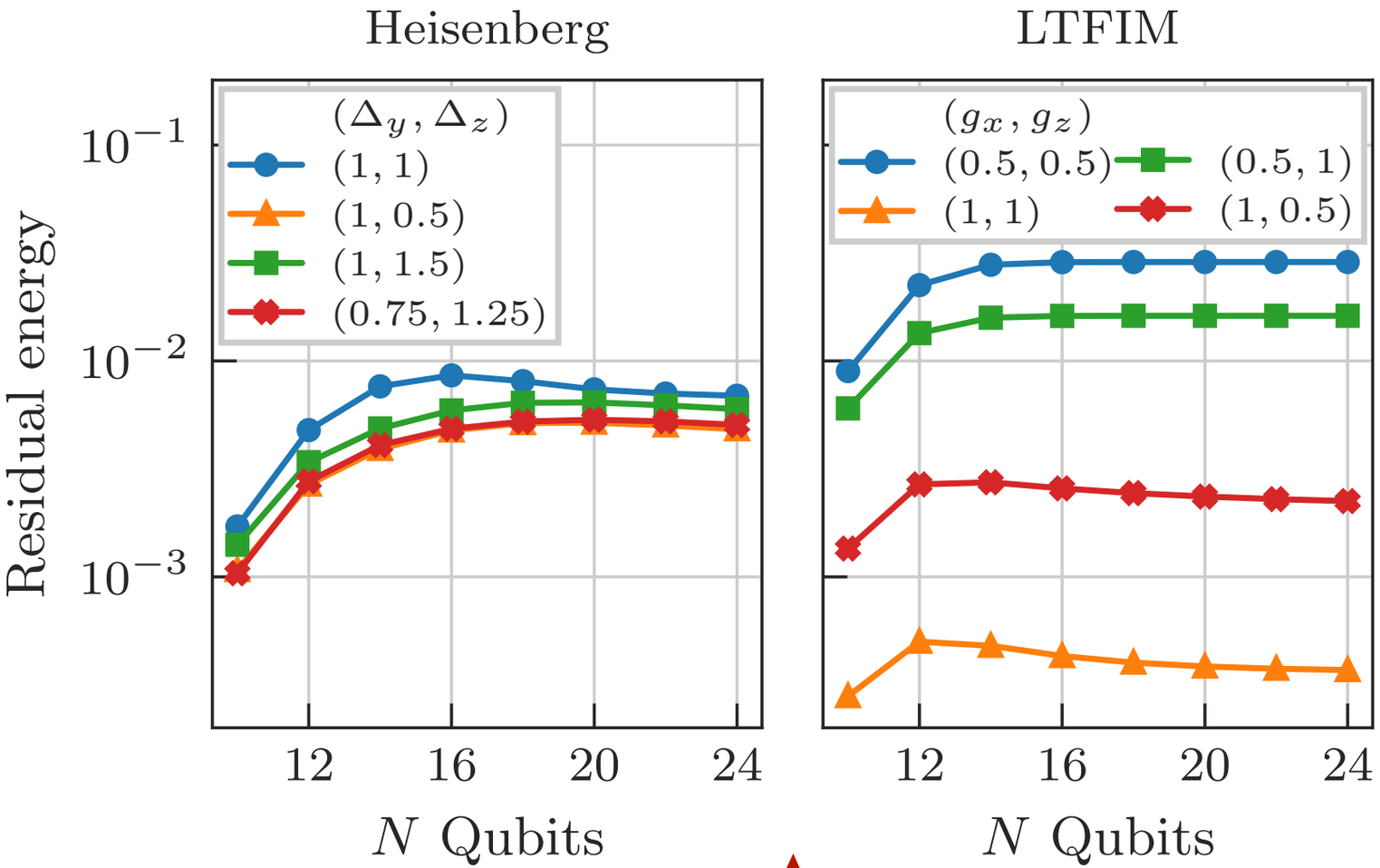
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Warm-start for the optimization when you go from P→P+1

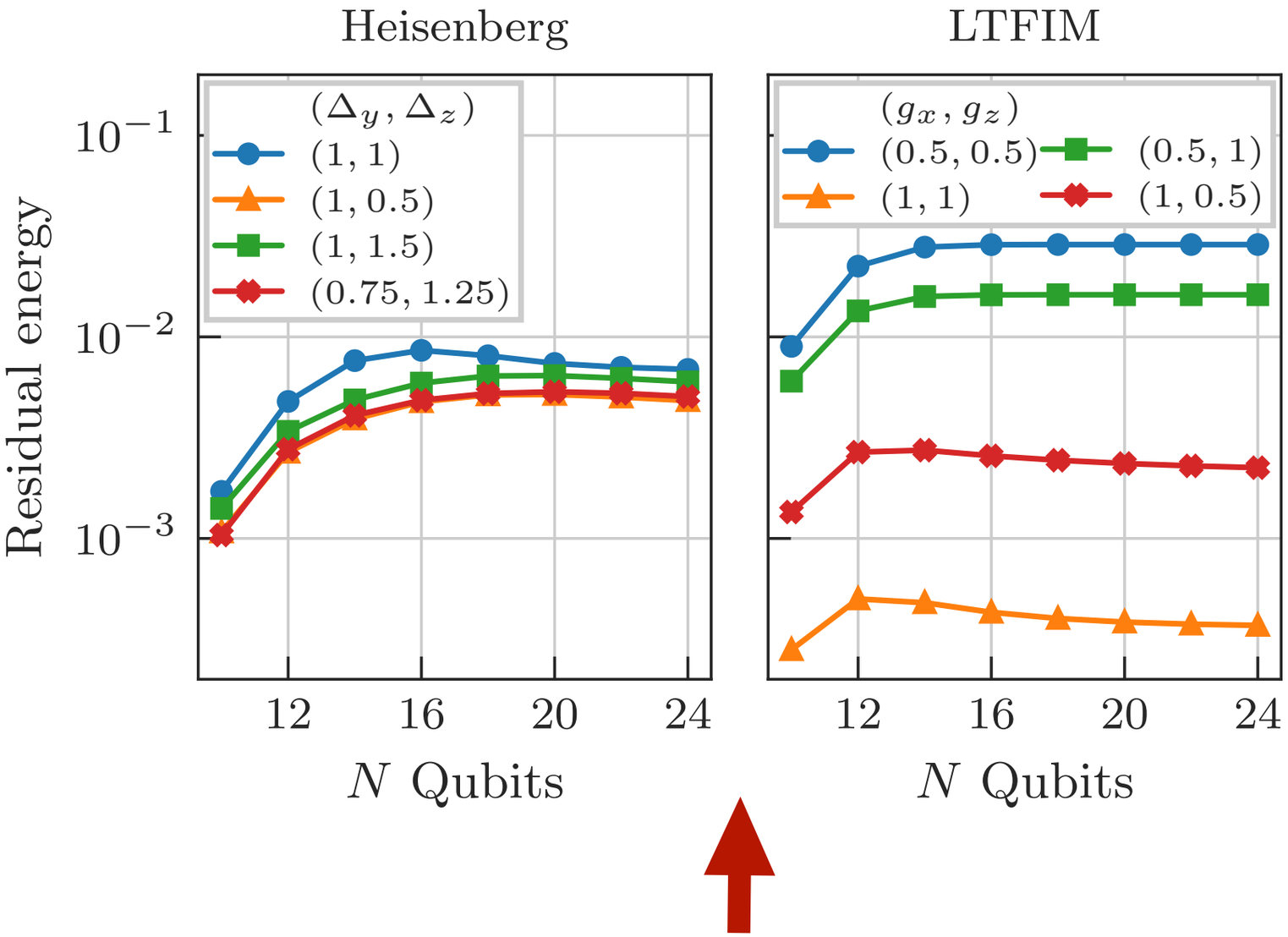


Transferability of solutions



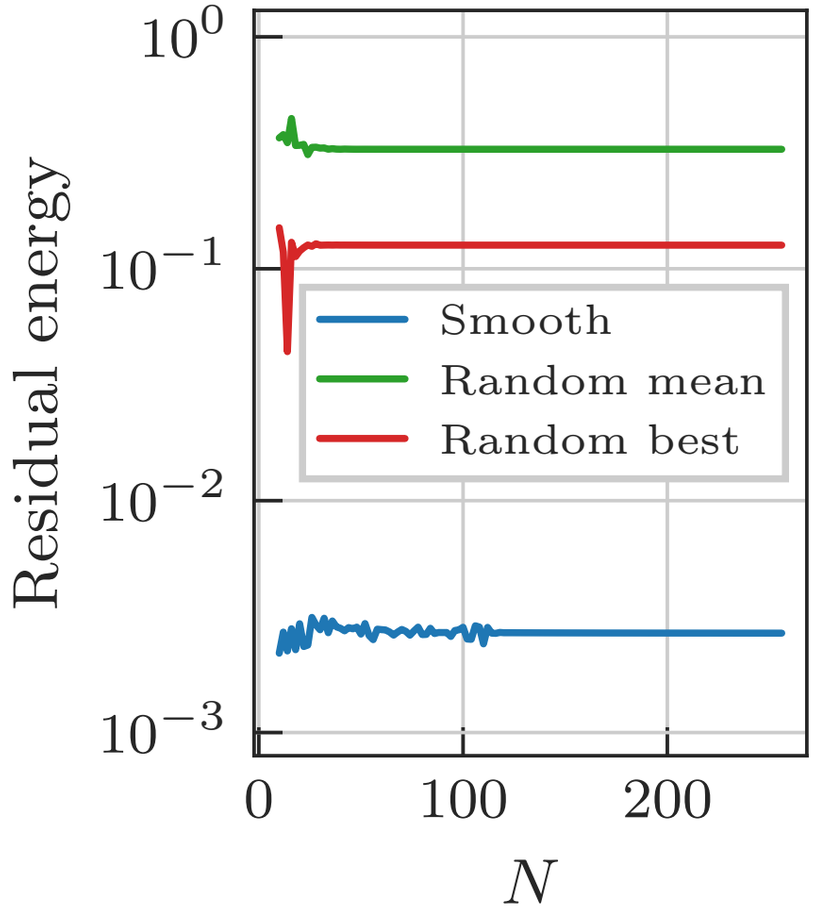
Warm-start when you go from $N_{GUESS} \rightarrow N_{TARGET}$

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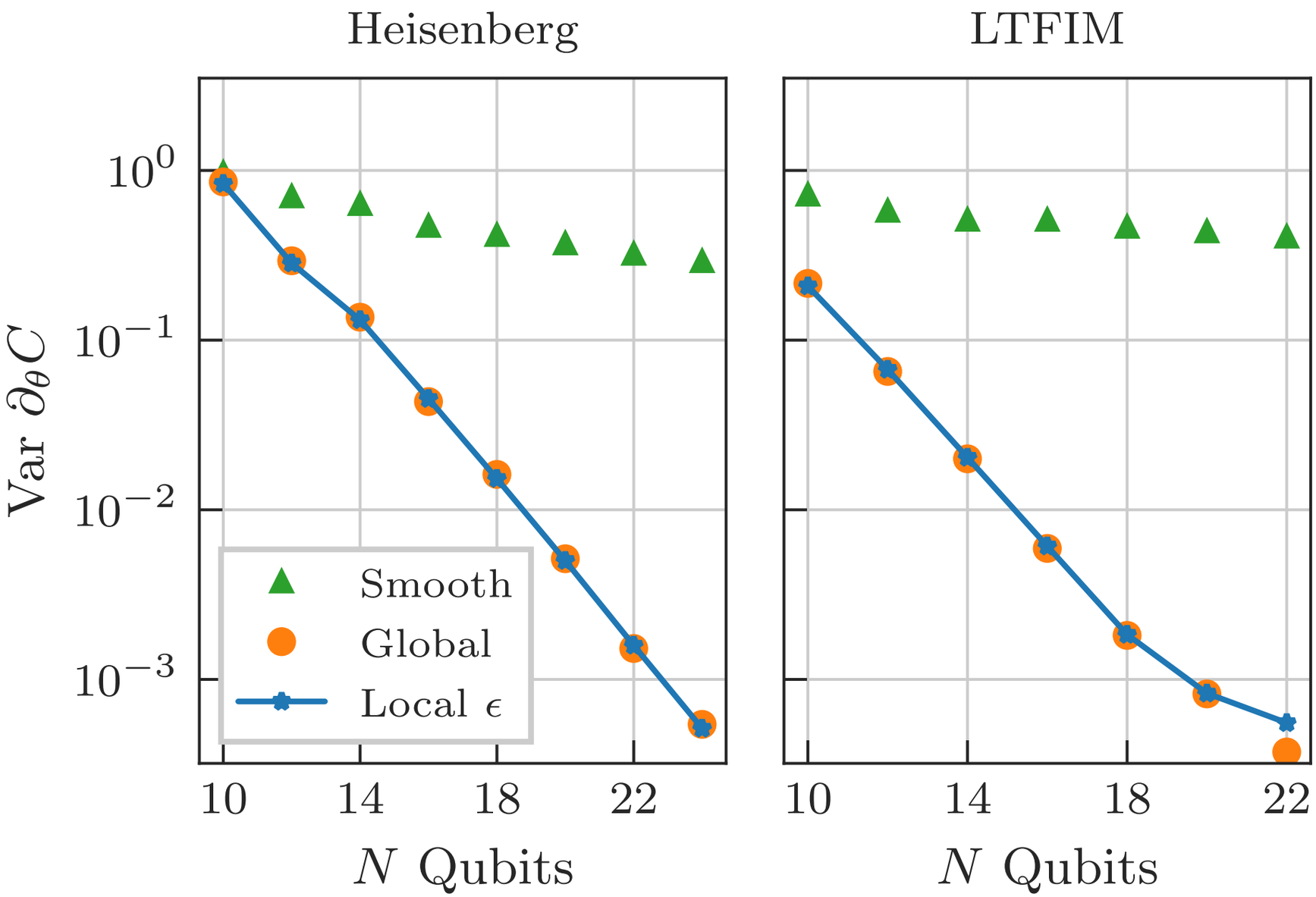


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Transferability also for large-N (TFIM)



The warm-start allows to avoid the flat region



SUMMARY

Solution transferability from small to larger system sizes, and from small to larger circuit depths.

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- 2D systems
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THANKS FOR YOUR ATTENTION!